Alexander Belik

Theoretical Physics | Quantum Thermodynamics | Scientific Computing

Dublin, Ireland
+353 85 256 4027

abelik1@outlook.com | linkedin.com/in/alexander-belik | github.com/ABelik1

PROFILE

Fourth-year Theoretical Physics student at Trinity College Dublin, focused on quantum thermodynamics, many-body dynamics, and research software.

I build reproducible Python and PyTorch tools that connect analytic models, simulation, machine learning, and experimental data workflows.

EDUCATION

B.A. Theoretical Physics

Trinity College Dublin, expected 2026

Quantum mechanics, statistical mechanics, quantum field theory, and computational physics.

Leaving Certificate

The Dublin Academy of Education, 2020–2022

CORE PHYSICS

Quantum thermodynamics
Many-body dynamics
Gibbs-preserving maps
Density matrices and channels
Semidefinite programming
Spectroscopy workflows
Computational chemistry

TECHNICAL TOOLKIT

Python, NumPy, SciPy
PyTorch, QuTiP, Optuna
Matplotlib, Mathematica
C/C++ simulations
Docker, SQLite
Git/GitHub, LaTeX

RESEARCH PRACTICE

Numerical simulation
Data analysis and diagnostics
Hyperparameter tuning
Reproducible pipelines
Technical writing

LANGUAGES

English – fluent
Russian – intermediate
Spanish – basic

RESEARCH EXPERIENCE

Local Thermality from Global Athermality — 2025–2026

Capstone thesis, supervisor: Felix Binder

- Studied locally thermal bipartite quantum states with Gibbs marginals, isolating correlations as the remaining thermodynamic resource.
- Characterized the locally thermal set as an affine slice of the positive semidefinite cone.
- Derived a nine-parameter two-qubit normal form and a Schur-complement positivity criterion.
- Analysed diagonal slices, locally thermal rays, commuting qutrit subclasses, and SDP convertibility tests using $D(\rho\|\Gamma) = I(A:B)_\rho$ on the locally thermal manifold.

Research Assistant, CRANN Institute — Jun–Sep 2025

Trinity College Dublin

- Designed Echo State Network reservoir models to learn quantum time-evolution data from simulated spin-chain observables.
- Built a PyTorch ESN toolkit with configurable reservoirs, mixed precision, ridge-regression solvers, Optuna tuning, and structured diagnostics.
- Paired the learning workflow with a Heisenberg-chain simulator and conservation checks for norm, energy, and magnetization.

Experimental / Computational Research Assistant — May–Sep 2024

Trinity College Dublin

- Created Python experiment-control and data-acquisition tooling for spectroscopy measurements on ferrofluids and liquid crystals.
- Automated structured logging and repeatable measurement workflows; related work: arXiv:2504.19633.

DFT Hamilton / PySCF Work — 2024–2025

Computational chemistry

- Built a Dockerized PySCF workflow for H2 RHF/DFT, open-shell Li UHF, molecular examples, spin-correlation tests, endpoint checks, and custom functional comparisons.

TEACHING AND TECHNICAL WORK

Teaching Assistant — Oct 2025–Present

Trinity College Dublin

- Lead weekly tutorials and problem-solving sessions aligned with lectures.
- Grade assignments and exams with clear, constructive feedback.

Software Engineer, EdTech — 2024–2025

Grinds360

- Built Python automation tools for content and workflow pipelines.
- Contributed to product features and UX iteration with technical and non-technical stakeholders.

SELECTED RESEARCH SOFTWARE

EchoState and Heisenberg Chain: physics-informed reservoir-computing toolkit for quantum dynamics prediction.

Spectroscopy Automation: experiment-control and acquisition tooling for cleaner lab data capture.

PySCF Environment: reproducible Docker workflow for computational chemistry experiments.